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Document Code: WI-01-MIC-04	COLIFORMS AND E. COLI IN WATER AND ICE		

1. OBJECTIVE

To determine the presence of coliform bacteria and *E. coli* in water and ice used in pre-processing and processing plants using the Most Probable Number (MPN) method.

2. SCOPE

This instruction is applicable in the determination of Coliform Bacteria and *E. coli* organism in water and ice used in pre-processing and processing plants. Water identities include Deepwell and water from approved government or private source.

3. RESPONSIBILITY

Laboratory Analyst.

4. APPARATUS/EQUIPMENT/CONSUMABLES

- 4.1 Alcohol lamps / Fireboy
- 4.2 Alcohol (70% solution isopropyl alcohol)
- 4.3 Autoclave
- 4.4 Spatula
- 4.5 Graduated cylinders (100 ml, Pyrex)
- 4.6 Dilution bottles with screw caps (160 ml, Pyrex)
- 4.7 Media bottles (250 and 500 ml, Pyrex)
- 4.8 Sterile Pipets (10 ml)
- 4.9 Sterile Pipettor
- 4.10 Sterile Pipet tips (100 – 1000 µl and 10 – 200 µl)
- 4.11 Sterile Petri dishes with cover (100x15mm, Pyrex)
- 4.12 Test Tubes with autoclavable caps (10 ml, 15 ml, and 34 ml, Pyrex)
- 4.13 Durham tubes (6x50 mm, Pyrex)
- 4.14 Test tube racks
- 4.15 Incubator maintained at 35 ± 1°C

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- 4.16 Covered water bath, with circulating system to maintain temperature of $45.5 \pm 0.2^{\circ}\text{C}$.
- 4.17 Inoculating loops
- 4.18 Inoculating needles
- 4.19 Biosafety Cabinet

5. MEDIA AND REAGENTS

5.1 All information regarding media preparation is placed in LF-W04-01, Reagent Preparation Logbook.

5.2 Butterfield's Phosphate-Buffered Dilution Water

Suspend 34 g of KH_2PO_4 to 500 milliliter of distilled water. Adjust pH to 7.2 with 1N NaOH. Bring volume to 1 liter with distilled water. Sterilize 15 min at 121°C . Store in a refrigerator.

Dilution Blanks

Take 1.25 ml of above stock solution and bring volume to 1 liter with distilled water. Dispense into bottles to 90 or 99 ± 1 ml. Sterilize 15 min at 121°C .

5.3 Pre-enrichment and Selective Media

5.3.1 Lauryl Tryptose (LST) Broth

Suspend 35.6 g in 1 liter of demineralized water. Dispense 10 ml portions in 20 ml test tubes with inverted Durham tubes. Autoclave 15 min at 121°C . Final pH, 6.8 ± 0.2 .

5.2.2. Brilliant-Green Lactose Bile (BGLB) Broth

Suspend 40 g in 1 liter of demineralized water. Dispense 10 ml volumes in 20 ml test tubes with inverted Durham tubes. Autoclave 15 min at 121°C . Final pH, 7.2 ± 0.1 .

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5.2.3 EC Medium

Suspend 37 g in a liter of demineralized water. Dispense 8 ml volumes in 15 ml test tubes with inverted Durham tubes. Autoclave 15 min at 121°C. Final pH, 6.9 ± 0.2 .

5.3 Differential Medium

5.3.1 Levine's Eosin Methylene Blue (L-EMB) Agar

Suspend 36 g of powder in a liter of distilled water. Mix thoroughly and dissolve completely by heating in a boiling water bath. Autoclave 15 min at 121°C. Final pH, 7.1 ± 0.2 .

5.4 Media for Biochemical Tests

5.4.1 Tryptone Broth (TB)

Suspend 10 g in a liter of distilled water. Dissolve completely. Dispense 5 ml volumes in 10 ml test tubes. Autoclave at 121°C for 15 minutes. Final pH, 6.9 ± 0.2 .

5.4.2 MR-VP Medium

Dissolve 17 g in 1 liter of demineralized water. Dispense 5 ml volumes in 10 ml test tubes. Autoclave at 121°C for 15 minutes. Final pH, 7.5 ± 0.2 .

5.4.3 Simmons Citrate Agar (SCA)

Suspend 24.28 g in a liter of demineralized water. Heat gently with occasional agitation. Boil for 1-2 minutes until agar dissolves. Fill 10 ml or 20 ml screw-cap tubes 1/3 full. Autoclave at 121°C for 15 minutes. Final pH, 6.7 ± 0.2 . Before medium solidifies, incline tubes to obtain 4-5 cm slants and 2-3 cm butts.

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5.5 Reagents

5.5.1 Erlich's Reagent

Dissolve 2.0 g of p-dimethylaminobenzaldehyde in 190 ml absolute ethyl alcohol. Add 40 ml concentrated HCl.

5.5.2 Kovac's Reagent

Dissolve 10 g of p-dimethylaminobenzaldehyde in 150 ml pure amyl or isoamyl alcohol and slowly add 50 ml concentrated HCl.

5.5.3 Methyl Red Indicator

Dissolve 0.1 g methyl red in 300 ml of 95% ethyl alcohol and dilute to 500 ml with distilled water.

5.5.4 α -naphthol solution

Dissolve 5 g purified α -naphthol in 100 ml absolute ethanol.

5.5.5 Potassium hydroxide solution

Dissolve 40 g potassium hydroxide in 100 ml distilled water.

6. PROCEDURE

6.1 Presumptive Test

6.1.1 Label 10 tubes of double strength LST (DSLST) with sample code.

6.1.2 Shake the sample well by moving the sample bottle 25 times in 30 cm arc or vortex mix for 7 s. Sampling bottle should have enough air space for effective shaking. If the bottle is completely full, aseptically pour out 20-30 ml of the water, replace cap and shake.

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- 6.1.3 Transfer 10 ml sample into each of 10 DSLST tubes. Mix gently.
- 6.1.4 Incubate the tubes at 35°C for 24 h.
- 6.1.5 After 24 h, examine the tubes for gas production. The presence of gas is a presumptive evidence of coliform bacteria. Reincubate the negative tubes for another 24 h.
- 6.1.6 Take note of the presumptive positive (gassing) tubes after 48 ± 2 h of incubation. Read the Most Probable Number (MPN) of coliform bacteria per 100 ml from the MPN Table 2. Record result.
- 6.1.7 Subject all presumptive positive tubes to confirmed test. It is also recommended to do confirmatory tests on turbid, non-gassing (negative) tubes as high levels of non-coliform bacteria can mask the lactose-fermenting activity of coliforms.
- 6.1.8 No further examination is needed for clear tubes that do not produce gas after 48 h of incubation.

6.2 Confirmed Test for Coliforms

- 6.2.1 Gently agitate each gassing DSLST tube and transfer a loopful or more of suspension into a BGLB broth tube.
- 6.2.2 Incubate the BGLB broth tubes at 35°C for 48 ± 2 h.
- 6.2.3 Examine the tubes for gas production and record the number of positive tubes.
- 6.2.4 Read the MPN of confirmed coliform bacteria per 100 ml sample from MPN Table 2. Record result.

6.3 Confirmed Test for Fecal Coliforms and Examination for Presumptive *E. coli*

- 6.3.1 Gently agitate each gassing DSLST tube and transfer a loopful of suspension into a tube of EC broth.
- 6.3.2 Incubate the EC broth tubes at 44.5 ± 0.2 °C for 48 ± 2 h.
- 6.3.3 Examine the tubes for gas production and record the number of positive tubes.

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- 6.3.4 Read the MPN of fecal coliform bacteria per 100 ml sample from MPN Table 2.
Record result.

6.4 Confirmed Test for *E. coli*

- 6.4.1 Streak a loopful of suspension from each gassing EC broth tube on L-EMB agar plates. Incubate the plates at 35°C for 18 – 24 h.
- 6.4.2 Examine plates for suspicious *E. coli* colonies i.e., dark-centered and flat, with or without greenish metallic sheen (see Figure 1).
- 6.4.3 Transfer up to 5 suspicious colonies from each L-EMB plate to PCA slants and incubate for 18-24 h at 35°C and use for further testing.

Note: Identification of any 1 of the 5 colonies as *E. coli* is sufficient to regard the EC broth tube as positive; hence, not all 5 isolates may need to be tested.

- 6.4.4 Perform Gram stain. All cultures appearing as Gram-negative, short rods should be tested for the IMViC reactions and also re-inoculated back into LST to confirm gas production.

6.5 IMViC Tests

6.5.1 Indole Production

Inoculate tube of Tryptone Broth and incubate at 35°C for 24 ± 2 h. Test for indole by adding 1 ml xylene, shake well and allow xylene to rise. Add 0.5 ml Erlich's reagent. Alternatively add 0.5 ml of Kovac's reagent directly to the TB culture. Appearance of distinct red color in upper layer indicates positive indole test.

6.5.2 Voges-Proskauer (VP)

Inoculate tube of MRVP broth and incubate for 48 ± 2 h at 35°C. Transfer 1 ml to 13 x 100 mm tube. Add 0.6 ml α-naphthol solution and 0.2 ml 40% KOH and shake. Add a few crystals of creatine. Shake and let stand for 2 h. Test is positive if eosin pink color develops.

6.5.3 Methyl Red

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After VP Test, incubate MRVP tube for an additional of 48 ± 2 h at 35°C. Add 5 drops of methyl red solution to each tube. Distinct red color is a positive test. Yellow is a negative reaction.

6.5.4 Citrate Test

Streak and stab tube of Simmons citrate agar slant with the isolate. Incubate for 96 h at 35°C. Growth on the medium with blue coloration is a positive reaction for citrate utilization.

6.5.5 Interpret the results of the IMViC tests according to the reactions of coliforms and other related microorganisms in Table 1.

Table 1. Reactions of *E. coli* and other related microorganisms in IMViC test.

Test	<i>Escherichia coli</i>	<i>Citrobacter freundii</i>	<i>Escherichia hermannii</i>	<i>Escherichia vulneris</i>	<i>Enterobacter aerogenes</i>	<i>Klebsiella</i>	<i>Pseudomonas</i>	<i>Aeromonas</i>
Indole	+	-	+	-	-	±	-	+
Methyl Red	+	+	±	+	-	-	-	-
Voges-Proskauer	-	-	-	-	+	-	-	+
Citrate	-	+	-	-	+	+	+	+

6.6 Quantitative Determination of *E. coli*

All cultures that give IMViC patterns of +++ are considered to be *E. coli*. Read Most Probable Number based on proportion of EC tubes in 3 successive dilutions that contain *E. coli* from Table 2.

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Table 2.MPN Index and 95% confidence limit for various combinations of positive and negative results when 10 ml portions are used.

No. of Tubes Giving Positive Reaction Out of 10 of 10 ml Each	MPN Index/100 ml	95% Confidence Limits (Approximate)	
		Lower	Upper
0	<1.1	0	3.0
1	1.1	0.03	5.9
2	2.2	0.26	8.1
3	3.6	0.69	10.6
4	5.1	1.3	13.4
5	6.9	2.1	16.8
6	9.2	3.1	21.1
7	12.0	4.3	27.1
8	16.1	5.9	36.8
9	23	8.1	59.5
10	>23.0	13.5	Infinite

7. REFERENCE

Bacteriological Analytical Manual, Edition 8, Revision A, 1998. Chapter 4. Authors: Peter Feng, Stephen D. Weagant (ret.), Michael A. Grant (dec.), William Burkhardt

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8. FORM

LF W01-01, Reagent and Media Preparation Logbook

LF W01-MIC-02, Analyst Worksheet for Bacteriological Examination of Water and Ice

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